THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 02

Full Syllabus (Chapters 1–14) | Class 10 Mathematics | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(d)** $2^2 \times 5 \times 7$	$140 = 4 \times 35$
2	**(c)** π	π is irrational
3	**(c)** Incorrect — a non-zero constant has degree 0	$7 = 7x^0$
4	**(a)** $-2, 5$	x-coordinates of intercepts
5	**(b)** Student A is correct	$5 - 3 = 2$
6	**(c)** 6	$\frac{2}{4} = \frac{3}{k} = \frac{5}{10}$
7	**(a)** 1	$49 - 48$
8	**(b)** $0, -5$	$x(x+5) = 0$
9	**(b)** 1.1	$1.7 - 0.6$
10	**(d)** 13th	$2 + 4(n-1) = 50$
11	**(d)** 7.5 m	$\frac{1.5}{2} = \frac{h}{10}$
12	**(c)** 60°	$\angle D = \angle A$
13	**(c)** $\frac{4}{5}$	Sides 3, 4, 5
14	**(d)** 2	One on each side
15	**(b)** 11 cm	$\frac{1}{8} \times 88$
16	**(a)** 38.5 cm²	A quadrant
17	**(a)** 1232 cm²	2×616
18	**(c)** class width	Step size
19	**(b)** Mode = 3 Median − 2 Mean	Standard relation
20	**(d)** $\frac{10}{13}$	$\frac{40}{52}$

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	$12^n = 2^{2n} \times 3^n$ has no prime factor 5, so it cannot end in 0 or 5.	2
22	$\frac{\alpha^2+\beta^2}{\alpha\beta} = \frac{9-2}{1} = 7$	1+1
23	$4(k+1)^2 - 4k^2 = 0 \Rightarrow 2k + 1 = 0 \Rightarrow k = -\frac{1}{2}$	1+1
24	$DB = 4.5$; $\frac{3}{4.5} = \frac{AE}{3} \Rightarrow AE = 2$ cm	1+1
25	Each side squared $= 8$ (e.g. $(1+\sqrt{3})^2 + (\sqrt{3}-1)^2 = 8$) ⇒ equilateral	2
26	$\cos\theta = \frac{15}{17}$, $\tan\theta = \frac{8}{15}$	1+1
27	$\frac{4}{\cos 45^\circ} = 4\sqrt{2} \approx 5.66$ m	2
28	$XC = XA$, $YC = YB$ ⇒ perimeter $= PA + PB = 20$ cm	2

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	Both pass through $(0,2), (6,0)$ — coincident ⇒ consistent, infinitely many solutions	2+1
30	$a = 250, h = 100$; $\Sigma fu = 6$ ⇒ $250 + \frac{600}{40} = 265$	1+1+1
31	$a_{n+1} - a_n = -5$ (constant) ⇒ AP; $a_1 = 4$, $S_{15} = \frac{15}{2}(8 - 70) = -465$	1+2
32	OT ⊥ PQ at R: $PR = 2.4$, $OR = 1.8$; $\triangle TRP \sim \triangle PRO$ ⇒ $TP = \frac{PR \cdot OP}{OR} = 4$ cm	1+1+1
33	$\frac{\theta}{360} \times 113.04 = 18.84 \Rightarrow \theta = 60^\circ$; arc $= \frac{1}{6} \times 37.68 = 6.28$ cm	2+1
34	$\frac{x}{10+x} = \frac{1}{3} \Rightarrow x = 5$	1+1+1
35	Girl: 75 m; boy: $50\sqrt{2} \approx 70.71$ m ⇒ girl's, by about 4.29 m	1+1+1
	OR: $75\sqrt{3} - 25\sqrt{3} = 50\sqrt{3} \approx 86.6$ m	1+1+1

36	$40 + \frac{13-8}{26-8-7} \times 20 = 49.09$	1+1+1
	OR: $5 + \frac{15-9}{30-9-9} \times 2 = 6$ — the student is correct	1+1+1

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	(i) 4, 3 (ii) both 5 (iii) rectangle (opposite sides equal, diagonals equal, adjacent sides unequal) (iv) $(1, 0.5)$	1+2+1+1
	OR: Ratio 3 : 4 ⇒ $P\left(-\frac{2}{7}, \frac{32}{7}\right)$ (3); mid-point $(0, 5)$, $AB = 2\sqrt{13}$ (2)	3+2
38	$(\sin A + \cos A) \cdot \frac{1}{\sin A \cos A} = \frac{1}{\cos A} + \frac{1}{\sin A}$ (3). $A = 45°$: $\sqrt{2} \times 2 = 2\sqrt{2} = \sqrt{2} + \sqrt{2}$ ✓ (2)	3+2
	OR: $\frac{\cos^2\theta}{\sin\theta} \cdot \frac{\sin^2\theta}{\cos\theta} \cdot \frac{1}{\sin\theta\cos\theta} = 1$	2+2+1
39	(i) $108\pi + 12\pi = 120\pi \approx 377.14$ cm³ (ii) $l = 5$: $72\pi + 9\pi + 15\pi = 96\pi \approx 301.71$ cm²	3+2
	OR: (i) $3.08 + 0.7187 \approx 3.799$ m³ ≈ 3798.7 L (ii) $8.8 + 3.08 = 11.88$ m² (iii) ₹297	2+2+1
40	$f_1 + f_2 = 9$; $32.5 = 30 + \frac{20-(f_1+14)}{12} \times 10 \Rightarrow f_1 = 3,\ f_2 = 6$	3+2
	OR: (i) $\frac{1940}{50} = 38.8$ (ii) Wrong — frequencies ignored (iii) cf 12, 30 ⇒ $20 + \frac{13}{18} \times 20 = 34.44$	2+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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